USN											

"24CS330"

JSS MAHAVIDYAPEETHA
JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

Department of Computer Science and Engineering

III Semester: CIE 3: Test II

Data Structures

Duration: 1hour 15 mins | Date: 08-01-2026 | Max. Marks: 30 Marks

NOTE: Question 1 is compulsory. Answer any one between question 2 and question 3.

Q.NO	CO	CD	PI	QUESTION	MARKS
1.	CO2	2.1.2	L3	a. Develop a C function to reverse the contents of single linked list of integer numbers.	05
		1.4.1	L2	b. Illustration with an example, memory representation of circular single linked list and double linked list.	05
	CO3	1.4.1	L1	c. Define recursion. Develop a recursive function for tower of Hanoi and trace the same for 3 disks.	05
2.	CO2	2.1.2	L3	a. Write a function called DELETE_APPEND (PH), where PH is a pointer to Header node in a circular single linked list of integers, which deletes the front node and appends it to the end of the list. Before writing the function, define node structure, create and initialize an empty list.	05
	CO3	2.1.2	L3	b. Construct the algorithm to convert a given infix expression to postfix form and trace the stacks content for the expression $A+(B*(C-D)E^F)/G$.	10
				OR	
3.	CO2	2.1.2	L3	a. Write a function called SEARCH-INSERT(PH, X) where PH is a pointer to Header node in a circular double linked list of integers and X is an integer value which searches for X and if it does not exist, appends it to the end of list.	05
	CO3	1.4.1	L2	b. Discuss priority queue.	05
		1.4.1	L3	c. A circular queue of size 7 contains four elements 22, 18, 15, 10 (inserted in the same sequence). The front index F = 4. Starting with this status, show the queue contents using circular representation while performing the following operations. After each operation, write the front item and rear item. Operations: i) Insert 12, 20 ii) Delete 2 items iii) Insert 25, 30, 35 iv) Delete 1 item v) Insert 40	05

CO#	Course Outcomes	Highest Level of Cognitive Domain
CO1	Using the concept of pointers to arrays, structures, and functions, perform different operations on linear list with array storage representation.	L3
CO2	Use the concept of linked list representation to implement various operations on Linear lists with different types of linked list.	L3
CO3	Solve the problems of conversion and evaluation of different types of expressions using stack data structure, comprehend the implementation of different types of Queues.	L3
CO4	Using tree data structures solve the problems of searching, sorting, expression conversion and expression evaluation.	L3
CO5	Applying graph traversal techniques solve the problems of testing connectivity and acyclicity in a graph.	L3

PI's	
1.4.1	Apply theory and principles of computer science and engineering to solve an engineering problem
2.1.2	Identify processes/modules/algorithms of a computer-based system and parameters to solve a problem

Blooms Taxonomy	
L1	Remembering
L2	Understanding
L3	Applying

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